Set-Valued Fan-Takahashi Inequalities with Set-Relations Based on Scalarization Methods

Ryota Iwamoto*[a] and Tamaki Tanaka[b]

5th, September, 2025

[a]Graduate school of Science and Technology, Niigata University, Niigata 950–2181, Japan,
E-mail: 1engtaiyanben@math.sc.niigata-u.ac.jp
[b]Faculty of Science, Niigata University, Niigata 950–2181, Japan,
E-mail: tamaki@math.sc.niigata-u.ac.jp



Contents

Introduction

Preliminaries

Main results

Scalarization functions

Characterizing functions of an ordering cone

Conclusion

Introduction

Background

- Georgiev and Tanaka [2] extended Fan-Takahashi minimax inequality to the form of set-valued maps.

- Kuwano, Tanaka, and Yamada [4] constructed the result of four types of set-valued minimax inequalities with set relations.

- **Our goal is to generalize the result of four types of set-valued minimax inequalities which is not related to the specific set-relations and scalarization functions.**

---

[2] Pando Gr. Georgiev and Tamaki Tanaka. “Vector-valued set-valued variants of Ky Fan’s inequality”. In: J. Nonlinear Convex Anal. 1.3 (2000), pp. 245–254.

[5] Issei Kuwano, Tamaki Tanaka, and Syuuji Yamada. “Unified scalarization for sets and set-valued Ky Fan minimax inequality”. In: J. Nonlinear Convex Anal. 11.3 (2010), pp. 513–525.

Background

- Georgiev and Tanaka [2] extended Fan-Takahashi minimax inequality to the form of set-valued maps.

- Kuwano, Tanaka, and Yamada [4] constructed the result of four types of set-valued minimax inequalities with set relations.

- **Our goal is to generalize the result of four types of set-valued minimax inequalities which is not related to the specific set-relations and scalarization functions.**

---

[2] Pando Gr. Georgiev and Tamaki Tanaka. "Vector-valued set-valued variants of Ky Fan's inequality". In: J. Nonlinear Convex Anal. 1.3 (2000), pp. 245–254.

[5] Issei Kuwano, Tamaki Tanaka, and Syuuji Yamada. "Unified scalarization for sets and set-valued Ky Fan minimax inequality". In: J. Nonlinear Convex Anal. 11.3 (2010), pp. 513–525.

Theorem (Fan-Takahashi [5])
Let X be a nonempty compact convex subset of a Hausdorff topological vector space and $f : X \times X \to \mathbb{R}$. If f satisfies the following conditions:

1. for each fixed $y \in X$, $f(\cdot, y)$ is lower semicontinuous,
2. for each fixed $x \in X$, $f(x, \cdot)$ is quasi concave,
3. $f(x, x) \leq 0$ for all $x \in X$,

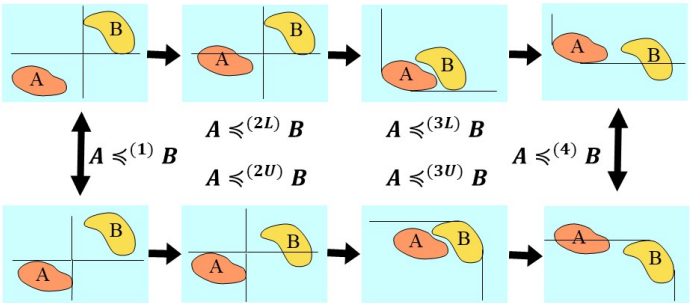
then there exists $\bar{x} \in X$ such that $f(\bar{x}, y) \leq 0$ for all $y \in X$.

Ordering and Set-relations

Let $(Y, \preccurlyeq)$ be an ordered space, generally.
$A, B \subset Y$: nonempty sets. $A \preccurlyeq^{(j)} B$ $(j = 1, 2L, 2U, 3L, 3U, 4)$

(1) $\forall a \in A, \forall b \in B, a \preccurlyeq b$ is defined below.

(2L) $\exists a \in A$ s.t. $\forall b \in B, a \preccurlyeq b$ **(3L)** $\forall b \in B,\ \exists a \in A$ s.t. $a \preccurlyeq b$



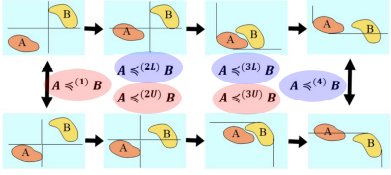
(2U) $\exists b \in B$ s.t. $\forall a \in A, a \preccurlyeq b$ **(3U)** $\forall a \in A, \exists b \in B$ s.t. $a \preccurlyeq b$

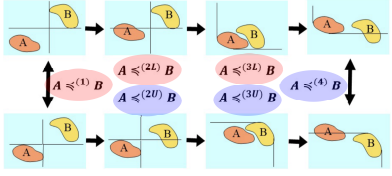
(4) $\exists a \in A, \exists b \in B$ s.t. $a \preccurlyeq b$

Lemma

Let Y be a real topological vector space, C a convex cone with $\operatorname{int} C \neq \emptyset$, and $A, \{\theta_Y\} \subset Y$. Then

1. $A \preccurlyeq_C^{(1)} \{\theta_Y\} \Leftrightarrow A \preccurlyeq_C^{(2U)} \{\theta_Y\} \Leftrightarrow A \preccurlyeq_C^{(3U)} \{\theta_Y\} \Leftrightarrow A \subset -C$,
2. $A \preccurlyeq_C^{(2L)} \{\theta_Y\} \Leftrightarrow A \preccurlyeq_C^{(3L)} \{\theta_Y\} \Leftrightarrow A \preccurlyeq_C^{(4)} \{\theta_Y\} \Leftrightarrow A \cap (-C) \neq \emptyset$,
3. $\{\theta_Y\} \not\preccurlyeq_{\operatorname{int} C}^{(1)} A \Leftrightarrow \{\theta_Y\} \not\preccurlyeq_{\operatorname{int} C}^{(2L)} A \Leftrightarrow \{\theta_Y\} \not\preccurlyeq_{\operatorname{int} C}^{(3L)} A \Leftrightarrow A \cap \operatorname{int} C = \emptyset$,
4. $\{\theta_Y\} \not\preccurlyeq_{\operatorname{int} C}^{(2U)} A \Leftrightarrow \{\theta_Y\} \not\preccurlyeq_{\operatorname{int} C}^{(3U)} A \Leftrightarrow \{\theta_Y\} \not\preccurlyeq_{\operatorname{int} C}^{(4)} A \Leftrightarrow A \not\subset \operatorname{int} C$.





Let $f : X \to \mathbb{R}$ and $F : X \to \mathcal{P}_0(Y)$.

$$f(\bar{x}, y) \leq 0 \quad \Longleftrightarrow \quad 0 \not\leq f(\bar{x}, y)$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$\boxed{F(\bar{x}, y) \preccurlyeq_C^{(j)} \{\theta_Y\}} \qquad \boxed{\{\theta_Y\} \not\preccurlyeq_{\operatorname{int} C}^{(j)} F(\bar{x}, y)}$$

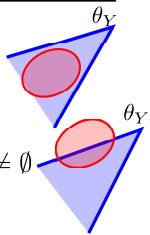
Case A1

$F(\bar{x}, y) \subset -C$

$\Downarrow$

Case A2

$F(\bar{x}, y) \cap (-C) \neq \emptyset$



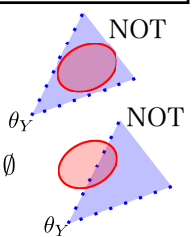
Case B1

$F(\bar{x}, y) \not\subset \operatorname{int} C$

$\Uparrow$

Case B2

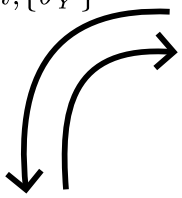
$F(\bar{x}, y) \cap \operatorname{int} C = \emptyset$

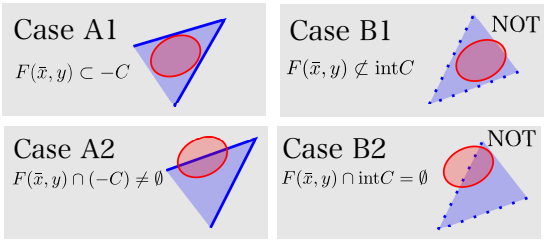


Four types of Set-Valued Minimax Inequalities with Set-relations

Assumption: F is (j) naturally quasi C-concave or (j) properly quasi C-concave

$S^{(j)}_{k,\{\theta_Y\}}$ or $I^{(j)}_{k,\{\theta_Y\}}$





proved by Kuwano, Tanaka, Yamada.

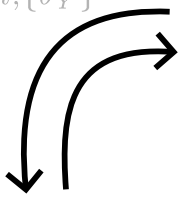
Fan-Takahashi minimax inequality

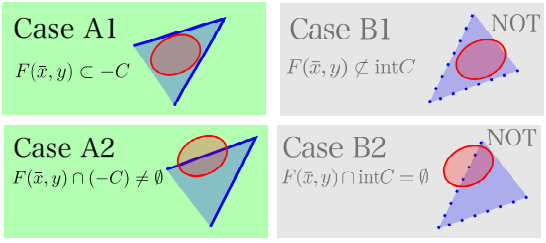
Four types of Set-Valued Minimax Inequalities with Set-relations

Assumption: F is (j) naturally quasi C-concave or (j) properly quasi C-concave

or $I^{(j)}_{k,\{\theta_Y\}}$

$S^{(j)}_{k,\{\theta_Y\}}$





proved by Kuwano, Tanaka, Yamada.

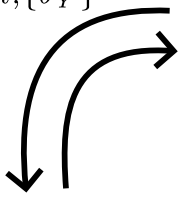
Fan-Takahashi minimax inequality

Four types of Set-Valued Minimax Inequalities with Set-relations

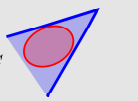
Assumption: F is (j) naturally quasi C-concave
or
(j) properly quasi C-concave

or $I^{(j)}_{k,\{\theta_Y\}}$

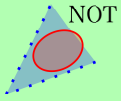
$S^{(j)}_{k,\{\theta_Y\}}$



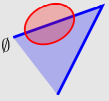
Case A1
$F(\bar{x}, y) \subset -C$



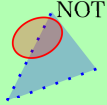
Case B1
$F(\bar{x}, y) \not\subset \mathrm{int}C$



Case A2
$F(\bar{x}, y) \cap (-C) \neq \emptyset$



Case B2
$F(\bar{x}, y) \cap \mathrm{int}C = \emptyset$



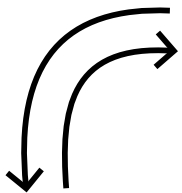
proved by Kuwano, Tanaka, Yamada.

Fan-Takahashi minimax inequality

Assumption: F is (j) naturally quasi C-concave

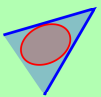
φ

(scalarization)



Case A1

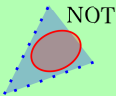
$F(\bar{x}, y) \subset -C$



Case B1

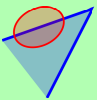
$F(\bar{x}, y) \not\subset \mathrm{int}C$

NOT



Case A2

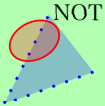
$F(\bar{x}, y) \cap (-C) \neq \emptyset$



Case B2

$F(\bar{x}, y) \cap \mathrm{int}C = \emptyset$

NOT



Fan-Takahashi minimax inequality

Preliminaries

Let X be a topological space, Y a real topological vector space, and θ_Y be a zero vector in Y. Define that $\mathcal{P}_0(Y)$ is the set of all nonempty subsets of Y. The sets of neighborhoods of $x \in X$ and $y \in Y$ is denoted by $\mathcal{N}_X(x)$ and $\mathcal{N}_Y(y)$, respectively.

Definition
For $A, B \in \mathcal{P}_0(Y)$, we define two binary relations on $\mathcal{P}_0(Y)$:

$$A \preccurlyeq_1 B \overset{\text{def}}{\Longleftrightarrow} A \cap B \neq \emptyset \quad \text{and} \quad A \preccurlyeq_2 B \overset{\text{def}}{\Longleftrightarrow} B \subset A.$$

Definition
A is said to be C-bounded if for each neighborhood U of θ_Y there exists $t > 0$ such that $A \subset tU + C$.

Definition

Let $f : Y \to \mathbb{R} \cup \{\pm\infty\}$ and $y_0 \in Y$. We say that f is lower semicontinuous (l.s.c. shortly) at y_0 if

$$\forall r < f(y_0), \exists V \in \mathcal{N}_Y(y_0) \text{ s.t. } r < f(y), \forall y \in V;$$

Definition [1]

Let $F : X \to \mathcal{P}_0(Y)$, $x_0 \in X$, $\preccurlyeq$ a binary relation on $\mathcal{P}_0(Y)$ and $C \subset Y$ a convex cone. We say that F is $(\preccurlyeq, C)$-continuous at x_0 if

$$\forall W \subset Y, W \text{ open}, W \preccurlyeq F(x_0), \exists V \in \mathcal{N}_X(x_0) \text{ s.t. } W + C \preccurlyeq F(x), \forall x \in V.$$

Remark

As special cases, $(\preccurlyeq_1, C)$-continuity and $(\preccurlyeq_2, C)$-continuity coincide with " C-lower continuity " and " C-upper continuity " for set-valued maps, respectively.

Definition [1]
Let $\varphi\colon \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$, $A_0 \in \mathcal{P}_0(Y)$, $\preccurlyeq$ a binary relation on $\mathcal{P}_0(Y)$, and C a convex cone in Y with $C \neq Y$. Then, we say that φ is $(\preccurlyeq, C)$-lower semicontinuous at A_0 if

$$\forall r < \varphi(A_0), \exists W \in \mathcal{P}_0(Y), W \text{ open, s.t. } W \preccurlyeq A_0 \text{ and } r > \varphi(A), \forall A \in U(W + C, \preccurlyeq);$$

where $U(V, \preccurlyeq) \coloneqq \{A \in \mathcal{P}_0(Y) \mid V \preccurlyeq A\}$.

Theorem [1]
Let $F\colon X \to \mathcal{P}_0(Y)$, $\varphi\colon \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$, $x_0 \in X$, $\preccurlyeq$ a binary relation on $\mathcal{P}_0(Y)$, and C a convex cone. If F is $(\preccurlyeq, C)$-continuous at x_0 and φ is $(\preccurlyeq, C)$-lower semicontinuous at $F(x_0)$, then $(\varphi \circ F)$ is lower semicontinuous at x_0.

Definition

Let X be a nonempty set, Y a real topological vector space, C a convex cone in Y, and $F\colon X \to \mathcal{P}_0(Y)$ a set-valued map.

1. F is called type (j) properly quasi C-concave if for each $x, y \in X$ and $\lambda \in (0,1)$,

$$F(x) \preccurlyeq_C^{(j)} F(\lambda x + (1-\lambda)y) \quad \text{or} \quad F(y) \preccurlyeq_C^{(j)} F(\lambda x + (1-\lambda)y)$$

2. F is called type (j) naturally quasi C-concave if for each $x, y \in X$ and $\lambda \in (0,1)$, there exists $\mu \in [0,1]$ such that

$$\mu F(x) + (1-\mu)F(y) \preccurlyeq_C^{(j)} F(\lambda x + (1-\lambda)y).$$

Remark

If F is type (j) properly quasi C-concave, then F is type (j) naturally quasi C-concave.

Definition [3]
Let $\mathcal{A} \subset \mathcal{P}_0(Y)$. $\mathcal{A}$ is said to be convex if for each $A_1, A_2 \in \mathcal{A}$ and $\lambda \in (0,1)$,

$$\lambda A_1 + (1-\lambda)A_2 \in \mathcal{A}.$$

Definition [3]
Let $\varphi \colon \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$. Then,

1. φ is concave if for $A, B \in \mathcal{P}_0(Y)$, $\varphi(\lambda A + (1-\lambda)B) \geq \lambda\varphi(A) + (1-\lambda)\varphi(B)$,
2. φ is quasi concave if for any $\alpha \in \mathbb{R}$, $\operatorname{lev}(\varphi, \geq, \alpha) \coloneqq \{A \in \mathcal{P}_0(Y) \mid \varphi(A) \geq \alpha\}$ is convex.

Remark
If φ is concave, then φ is quasi concave.

Preliminaries (Monotonicity)

Definition

Let C be a convex cone in Y. A scalarization function φ is $(\preccurlyeq_C^{(j)})$-monotone if for any $A, B \in \mathcal{P}_0(Y)$ with $A \preccurlyeq_C^{(j)} B$, $\varphi(A) \leq \varphi(B)$.

Definition

Let C be a convex cone in Y. A scalarization function φ is $(\preccurlyeq_{\operatorname{int} C}^{(j)})$-monotone if for any $A, B \in \mathcal{P}_0(Y)$ with $A \preccurlyeq_{\operatorname{int} C}^{(j)} B$, $\varphi(A) < \varphi(B)$.

Proposition

Let φ be $(\preccurlyeq_C^{(j)})$-monotone and quasi concave. If F is type (j) naturally quasi C-concave, then $(\varphi \circ F)$ is quasi concave.

Proposition

Let φ be $(\preccurlyeq_{\operatorname{int} C}^{(j)})$-monotone and quasi concave. If F is type (j) naturally quasi C-concave, then $(\varphi \circ F)$ is quasi concave.

Main results

Specific Scalarization Function (i)

Let $\varphi : \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$, $\preccurlyeq$ a binary relation on $\mathcal{P}_0(Y)$, and $C' \subset Y$ a convex cone. In order to generalize four types of set-valued minimax inequalities [4], we provide a new class of scalarization functions which satisfy;

(sc) φ is $(\preccurlyeq, C')$-lower semicontinuous;

(qv) φ is quasi concave.

In addition, we give necessary conditions between values of scalarization function and set-relations as follows;

Specific Scalarization Function (ii)

(i) φ is $(\preccurlyeq_C^{(j)})$-monotone,

(ii) $\varphi(A) \leq 0 \Rightarrow A \preccurlyeq_C^{(j)} \{\theta_Y\}$ for any $A \in \mathcal{P}_0(Y)$,

(i)' φ is $(\preccurlyeq_{\mathrm{int}\, C}^{(j)})$-monotone,

(ii)' $\varphi(A) > 0 \Rightarrow \{\theta_Y\} \preccurlyeq_{\mathrm{int}\, C}^{(j)} A$ for any $A \in \mathcal{P}_0(Y)$, and

(iii) $\varphi(\{\theta_Y\}) \leq 0$.

Supposed that φ satisfies conditions (sc) (qv), (i), (ii), and (iii), it is written as $\varphi \in \Phi(\preccurlyeq_C^{(j)}, \preccurlyeq, C')$. On the other hand, if φ satisfies conditions (sc) (qv), (i)', (ii)', and (iii), we write the notation as $\varphi \in \Phi(\preccurlyeq_{\mathrm{int}\, C}^{(j)}, \preccurlyeq, C')$.

Theorem

Let X be a nonempty compact convex subset of a Hausdorff topological vector space, Y a real topological vector space, $\preccurlyeq$ a binary relation on $\mathcal{P}(Y)$, C a convex cone in Y, C' a convex cone in Y, $\varphi\colon \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$, and $F\colon X \times X \to \mathcal{P}_0(Y)$. For the scaralization function $\varphi \in \Phi(\preccurlyeq^{(j)}_{\operatorname{int} C}, \preccurlyeq, C')$, if F satisfies the following conditions:

1. $(\varphi \circ F)(x, y) \in \mathbb{R}$ for all $x, y \in X$,
2. for each fixed $y \in X$, $F(\cdot, y)$ is $(\preccurlyeq, C')$-continuous,
3. for each fixed $x \in X$, $F(x, \cdot)$ is j-naturally quasi C-concave,
4. for all $x \in X$, $\{\theta_Y\} \not\preccurlyeq^{(j)}_{\operatorname{int} C} F(x, x)$,

then there exists $\bar{x} \in X$ such that $\{\theta_Y\} \not\preccurlyeq^{(j)}_{\operatorname{int} C} F(\bar{x}, y)$ for all $y \in X$.

Scalarization functions

Definition [4]

Let C be a proper closed convex cone in Y with $\operatorname{int} C \neq \emptyset$, $V, V' \in \mathcal{P}(Y)\backslash\{\emptyset\}$, and direction $k \in \operatorname{int} C$. For each $j = (3U), (3L)$, $I^{(j)}_{k,V'}(V) \colon \mathcal{P}(Y) \to \mathbb{R} \cup \{\pm\infty\}$ are defined by

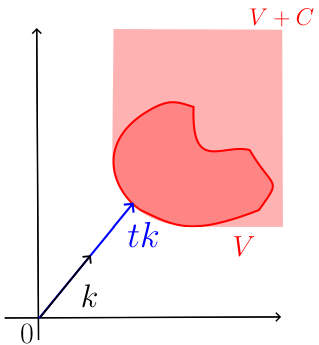
$$I^{(j)}_{k,V'}(V) := \inf\{t \in \mathbb{R} \mid V \preccurlyeq^{(j)}_C (tk + V')\}.$$

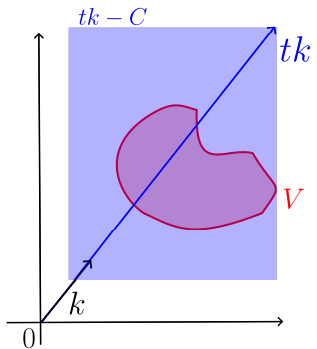
Example

$$I^{(3L)}_{k,\{\theta_Y\}}(V) := \inf\{t \in \mathbb{R} \mid V \preccurlyeq^{(3L)}_C (tk + V')\} = \inf\{t \in \mathbb{R} \mid tk \subset V + C\},$$
$$I^{(3U)}_{k,\{\theta_Y\}}(V) := \inf\{t \in \mathbb{R} \mid V \preccurlyeq^{(3U)}_C (tk + V')\} = \inf\{t \in \mathbb{R} \mid V \subset tk - C\}.$$

Example

$$I^{(3L)}_{k,\{\theta_Y\}}(V) := \inf\{t \in \mathbb{R} \mid V \preccurlyeq^{(3L)}_C \{tk\}\} = \inf\{t \in \mathbb{R} \mid tk \subset V + C\},$$
$$I^{(3U)}_{k,\{\theta_Y\}}(V) := \inf\{t \in \mathbb{R} \mid V \preccurlyeq^{(3U)}_C \{tk\}\} = \inf\{t \in \mathbb{R} \mid V \subset tk - C\}.$$





Definition (Hiriart-Urruty Oriented distance) [6]

Let Y be a real normed vector space. For a set $A \subset Y$, let the oriented distance function $\Delta_A \colon Y \to \mathbb{R} \cup \{\pm\infty\}$ be defined by

$$\Delta_A(y) \coloneqq d_A(y) - d_{Y \setminus A}(y),$$

$d_A(y) = \inf\{\|y - z\| \mid z \in A\}$, $d_\emptyset(y) \coloneqq +\infty$, and $\|y\|$ denotes the norm of y in Y.

Definition [6]

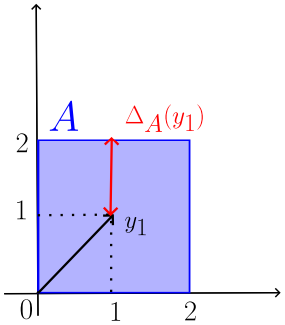
For the set $A \in Y$, let the functions $\mathcal{D}_A^+ \colon \mathcal{P}(Y) \to \mathbb{R} \cup \{\pm\infty\}$ and $\mathcal{D}_A^- \colon \mathcal{P}(Y) \to \mathbb{R} \cup \{\pm\infty\}$ be defined as

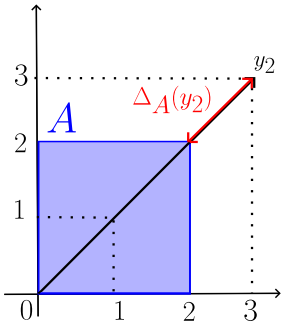
$$\mathcal{D}_A^+(B) \coloneqq \sup\{\Delta_A(b) \mid b \in B\},\ B \in \mathcal{P}(Y),$$
$$\mathcal{D}_A^-(B) \coloneqq \inf\{-\Delta_A(b) \mid b \in B\} = -\mathcal{D}_A^+(B),\ B \in \mathcal{P}(Y).$$

Example

Let $Y = \mathbb{R}^2$, $A = [0,2] \times [0.2]$, $y_1 = (1,1)$, and $y_2 = (3,3)$. Then,

$$\Delta_A(y_1) = d_A(y_1) - d_{Y \setminus A}(y_1) = 0 - 1 = -1,$$
$$\Delta_A(y_2) = d_A(y_2) - d_{Y \setminus A}(y_2) = 1 - 0 = \sqrt{2}.$$





Characterizing functions of an ordering cone

Characterizing functions

Definition

Let C be a convex cone in Y and $f : Y \to \mathbb{R} \cup \{\pm\infty\}$. We say that f is a characterizing function of the ordering cone $-C$ if

$$f(y) \leq 0 \Leftrightarrow y \in -C \quad \text{for all } y \in Y.$$

Lemma

Let C be a closed convex cone in Y and $f : Y \to \mathbb{R} \cup \{\pm\infty\}$ a characterizing function of $-C$. Define $\varphi_s : \mathcal{P}_0(Y) \to \mathbb{R} \cup \{\pm\infty\}$ by

$$\varphi_s(A) = \sup_{a \in A} f(a) \quad \text{for all } A \in \mathcal{P}_0(Y).$$

If f satisfies that for all $a, b \in Y$ with $a \leq_C b$, $f(a) \leq f(b)$, then φ_s is an element satisfying assumptions (i)–(iii) of type (3U).

Conclusion

Conclusion

- We introduce the background and the basic notion.
- We gave a new result of set-valued Fan-Takahashi inequalities via scalarization methods.
- Next step is to check other scalarization functions, in particular marginal functions, to satisfy new assumption we suggest.

References

[1] Premyuda Dechboon. **"Inheritance properties on cone continuity for set-valued maps via scalarization".** PhD thesis. 新潟大学, 2022. URL: https://ci.nii.ac.jp/naid/500001551932.

[2] Pando Gr. Georgiev and Tamaki Tanaka. **"Vector-valued set-valued variants of Ky Fan's inequality".** In: *J. Nonlinear Convex Anal.* 1.3 (2000), pp. 245–254. ISSN: 1345-4773,1880-5221.

[3] Shogo Kobayashi, Yutaka Saito, and Tamaki Tanaka. **"Convexity for compositions of set-valued map and monotone scalarizing function".** In: *Pac. J. Optim.* 12.1 (2016), pp. 43–54. ISSN: 1348-9151,1349-8169.

[4] Issei Kuwano, Tamaki Tanaka, and Syuuji Yamada. **"Unified scalarization for sets and set-valued Ky Fan minimax inequality".** In: *J. Nonlinear Convex Anal.* 11.3 (2010), pp. 513–525. ISSN: 1345-4773,1880-5221.

[5] Wataru Takahashi. **"Nonlinear variational inequalities and fixed point theorems".** In: *J. Math. Soc. Japan* 28.1 (1976), pp. 168–181. ISSN: 0025-5645,1881-1167. DOI: 10.2969/jmsj/02810168. URL: https://doi.org/10.2969/jmsj/02810168.

[6] YD Xu and SJ Li. **"A new nonlinear scalarization function and applications".** In: *Optimization* 65.1 (2016), pp. 207–231.

Thank you for your listening!

Theme

Get the source of this theme and the demo presentation from

`github.com/matze/mtheme`

The theme *itself* is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.

